

AI ASSISTED CODING

LAB TEST-2

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BATCH -18

K.1

PROMPT:-

I need a matrix rotation helper for urban public transit.
Rotate a matrix 90 degrees clockwise in place and print the result

Sample input:-

```
[[1,2,3],[4,5,6],[7,8,9]]
```

Sample output:-

```
[[7,4,1],[8,5,2],[9,6,3]]
```

Code:-

```
1 def rotate_matrix_90_clockwise(matrix):
2     """
3     Rotates an NxN matrix 90 degrees clockwise in-place using transpose + reverse each row.
4     Args:
5         matrix (list of list of int): NxN matrix to rotate
6     Returns:
7         None (modifies matrix in-place)
8     """
9
10    n = len(matrix)
11    # Transpose the matrix
12    for i in range(n):
13        for j in range(1, n):
14            matrix[i][j], matrix[j][i] = matrix[j][i], matrix[i][j]
15    # Reverse each row
16    for row in matrix:
17        row.reverse()
18
19    # --- Demo ---
20    if __name__ == "__main__":
21        user_matrix = [
22            [1, 2, 3],
23            [4, 5, 6],
24            [7, 8, 9]
25        ]
26        print("Original matrix:")
27        for row in user_matrix:
28            print(row)
29        rotate_matrix_90_clockwise(user_matrix)
30        print("\nRotated matrix (each column shown as a list):")
31        n = len(user_matrix)
32        for col in range(n):
33            print([user_matrix[row][col] for row in range(n)])
```

Output:-

```
PS C:\Users\bathu> & C:\Users\bathu\AppData\Local\Programs\Python\python313\python.exe c:\Users\bathu\python\first.py
Original matrix:
[1, 2, 3]
[4, 5, 6]
[7, 8, 9]
Rotated matrix (each column shown as a list):
[7, 8, 9]
[4, 5, 6]
[1, 2, 3]
```

K.2

PROMPT:-

I need a python code which Return (added, removed) between old and new lists preserving order

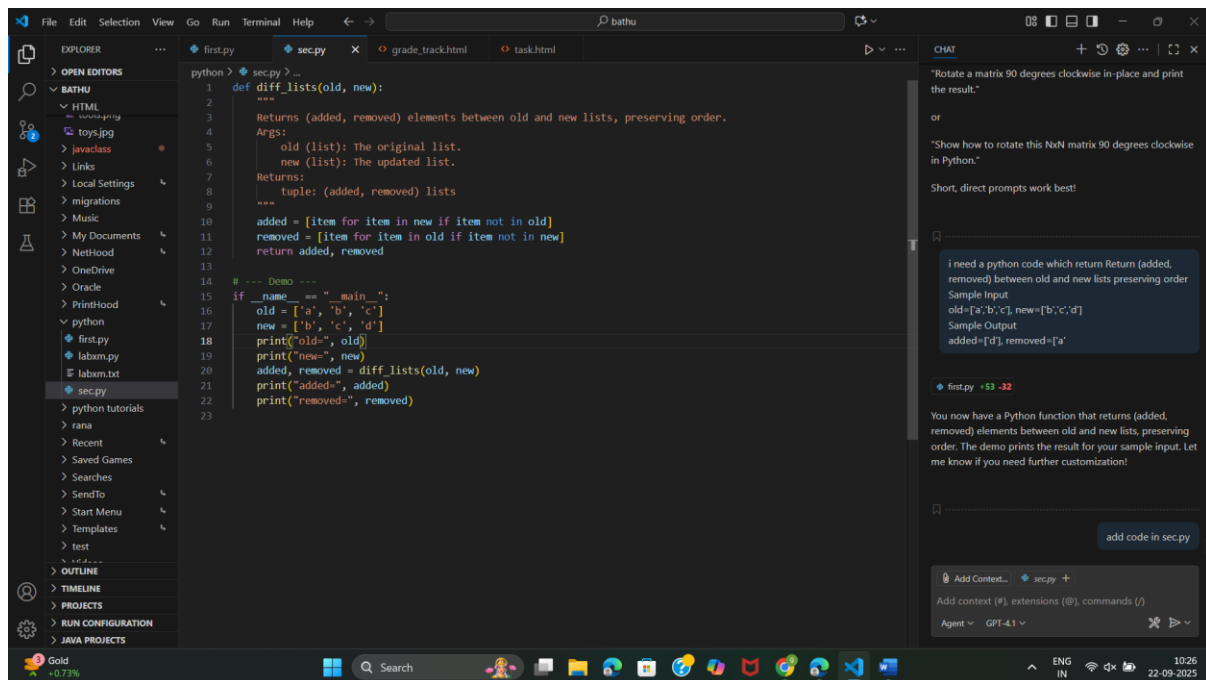
Sample Input

old=['a','b','c'], new=['b','c','d']

Sample Output

added=['d'], removed=['a']

code:-



The screenshot shows a Visual Studio Code editor with a Python file named `sec.py` open. The code defines a function `diff_lists` that takes two lists, `old` and `new`, and returns a tuple of lists containing elements added and removed. A demo is included in the `__main__` block, using the sample input and output provided. To the right of the editor is a chat window with a prompt and a response.

```
python > sec.py > --
1 def diff_lists(old, new):
2     """
3     Returns (added, removed) elements between old and new lists, preserving order.
4     Args:
5         old (list): The original list.
6         new (list): The updated list.
7     Returns:
8         tuple: (added, removed) lists
9     """
10    added = [item for item in new if item not in old]
11    removed = [item for item in old if item not in new]
12    return added, removed
13
14    # --- Demo ---
15    if __name__ == "__main__":
16        old = ['a', 'b', 'c']
17        new = ['b', 'c', 'd']
18        print("old=", old)
19        print("new=", new)
20        added, removed = diff_lists(old, new)
21        print("added=", added)
22        print("removed=", removed)
23
```

CHAT

"Rotate a matrix 90 degrees clockwise in-place and print the result."

or

"Show how to rotate this NxN matrix 90 degrees clockwise in Python."

Short, direct prompts work best!

i need a python code which return Return (added, removed) between old and new lists preserving order

Sample Input
old=['a','b','c'], new=['b','c','d']

Sample Output
added=['d'], removed=['a']

first.py +53 -32

You now have a Python function that returns (added, removed) elements between old and new lists, preserving order. The demo prints the result for your sample input. Let me know if you need further customization!

add code in sec.py

Add Context... sec.py +

Add context (#), extensions (@), commands (/)

Agent - GPT-4.1

10:26
22-09-2025

output:-

